

Sun-Glizide 30MR Tablet (Gliclazide)

INGREDIENT

Each Tablet contains:
Gliclazide.....30 mg

CLINICAL PHARMACOLOGY:

Pharmacodynamic:

Gliclazide is a hypoglycaemic sulphonylurea oral antidiabetic active substance differing from other related compounds by an N-containing heterocyclic ring with an endocyclic bond.

Gliclazide reduces blood glucose levels by stimulating insulin secretion from the β -cells of the islets of Langerhans. Increase in postprandial insulin and C-peptide secretion persists after two years of treatment.

In addition to these metabolic properties, gliclazide has haemovascular properties.

Effects on insulin release

In type 2 diabetics, gliclazide restores the first peak of insulin secretion in response to glucose and increases the second phase of insulin secretion. A significant increase in insulin response is seen in response to stimulation induced by a meal or glucose.

Haemovascular properties

Gliclazide decreases microthrombosis by two mechanisms which may be involved in complications of diabetes:

- a partial inhibition of platelet aggregation and adhesion, with a decrease in the markers of platelet activation (beta thromboglobulin, thromboxane B_2).
- an action on the vascular endothelium fibrinolytic activity with an increase in tPA activity.

Pharmacokinetics:

Plasma levels increase progressively during the first 6 hours, reaching a plateau which is maintained from the sixth to the twelfth hour after administration.

Intra-individual variability is low.

Gliclazide is completely absorbed. Food intake does not affect the rate or degree of absorption.

The relationship between the dose administered ranging up to 120 mg and the area under the concentration time curve is linear.

Plasma protein binding is approximately 95%.

Gliclazide is mainly metabolised in the liver and excreted in the urine: less than 1% of the unchanged form is found in the urine. No active metabolites have been detected in plasma.

The elimination half-life of gliclazide varies between 12 and 20 hours.

The volume of distribution is around 30 litres.

No clinically significant changes in pharmacokinetic parameters have been observed in elderly patients.

A single daily dose of Sun-Glizide 30MR Tablet maintains effective gliclazide plasma concentrations over 24 hours.

INDICATIONS AND USAGE:

Non-insulin-dependent diabetes mellitus (type 2), in association with dietary measures and with exercise, when these measures alone are not sufficient.

CONTRAINDICATIONS:

This medicine should not be used in following cases:

- known hypersensitivity or allergy to gliclazide, other sulphonylureas, sulfonamides, or to any of the excipients of the product;
- unstable and/or insulin-dependent diabetes mellitus, particularly juvenile diabetes, diabetic ketoacidosis, diabetic coma;
- during stress conditions such as serious infection, trauma or surgery;
- in the presence of liver disease or renal impairment;
- pregnancy or lactation.

WARNINGS:

Use of gliclazide modified-release must be considered as treatment in addition to proper dietary regimen and not as substitute for diet.

Patients over a period of time, may become progressively less responsive to therapy with oral hypoglycemic agents because of worsening of their diabetic state. If a loss of adequate blood glucose-lowering response to gliclazide modified-release is detected, the drug should be discontinued.

PRECAUTIONS:

Patient Selection and Follow-up: Careful selection of patients is important. It is imperative that there be rigid attention to diet, careful adjustment of dosage and instruction of the patient on hypoglycemic reactions, their recognition, remedies and control as well as regular, thorough medical follow-up.

Since the effects of oral hypoglycemic agents on the vascular changes and other long-term sequelae of diabetes mellitus are not fully known, patients receiving such drugs must be closely observed for both short- and long-term complications. Periodic assessment of cardiovascular, ophthalmic, renal and hepatic status is advisable.

In patients stabilized on gliclazide therapy, loss of blood sugar control may occur in cases of acute intercurrent disease or in stressful situations such as trauma or surgery. Under these conditions, discontinuation of gliclazide modified-release and administration of insulin should be considered.

The efficacy of any oral antidiabetic agent, including gliclazide, in reducing glucose to the desired level decreases over a long period of time in many patients: this may be due to progression in the severity of the diabetes, or to a reduced response to treatment. This phenomenon is known as secondary failure and should be distinguished from primary failure, when the drug is ineffective when prescribed as first-line treatment. Adequate dose adjustment and compliance with dietary measures should be considered before classifying the patient as secondary failure.

The metabolism and excretion of sulphonylureas including gliclazide modified-release, may be slowed in patients with impaired renal and/or hepatic function. If hypoglycemia should occur in such patients, it may be prolonged and appropriate management should be instituted. In such patients, blood and urine glucose should be regularly monitored. Measurements of glycated hemoglobin levels may also be useful.

Hypoglycemic Reactions: As with other sulphonylurea drugs, manifestations of hypoglycemia including dizziness, lack of energy, drowsiness, headache and sweating have been observed and weakness, nervousness, shakiness and paresthesia have also been reported. Severe hypoglycemia can be induced by all sulphonylurea drugs. Particularly susceptible are elderly subjects, patients with impaired hepatic or renal function, those who are debilitated or malnourished and patients with primary or secondary adrenal insufficiency. Hypoglycemia may be difficult to recognize in elderly patients and in patients receiving β -blockers.

This treatment should only be prescribed if the patient is likely to have a regular food intake (including breakfast). It is important to have a regular carbohydrate intake due to the increased risk of hypoglycemia if a meal is taken late, if an inadequate amount of food is consumed or if the food is low in carbohydrates. Hypoglycemia is more likely to occur during periods of low-calorie diet, following prolonged or strenuous exercise, following alcohol intake or during the administration of a combination of hypoglycemic agents.

Children: Safety and effectiveness of gliclazide modified-release in children have not been established. Gliclazide modified-release is therefore not recommended for children and adolescents.

DRUG INTERACTIONS:

1) The following products are likely to increase the risk of hypoglycaemia

Contra-indicated combination

- **Miconazole** (systemic route, oromucosal gel): increases the hypoglycaemic effect with possible onset of hypoglycaemic symptoms, or even coma.

Combinations which are not recommended

- Phenylbutazone (systemic route): increases the hypoglycaemic effect of sulphonylureas (displaces their binding to plasma proteins and/or reduces their elimination).

- It is preferable to use a different anti-inflammatory agent, or else to warn the patient and emphasise the importance of self-monitoring. Where necessary, adjust the dose during and after treatment with the anti-inflammatory agent.

- **Alcohol:** increases the hypoglycaemic reaction (by inhibiting compensatory reactions) that can lead to the onset of hypoglycaemic coma. Avoid alcohol or medicines containing alcohol.

Combinations requiring precautions for use

Potential of the blood glucose lowering effect and thus, in some instances, hypoglycaemia may occur when one of the following drugs is taken, for example: Other antidiabetic agents (insulins, acarbose, biguanides), beta-blockers, fluconazole, angiotensin converting enzyme inhibitors (captopril, enalapril), H2-receptor antagonists, MAOIs, sulfonamides, and nonsteroidal anti-inflammatory agents.

2) The following products may cause an increase in blood glucose levels

Combination which is not recommended

- **Danazol:** diabetogenic effect of danazol.

If the use of this active substance cannot be avoided, warn the patient and emphasise the importance of urine and blood glucose monitoring. It may be necessary to adjust the dose of the antidiabetic agent during and after treatment with danazol.

Combinations requiring precautions during use

- **Chlorpromazine** (neuroleptic agent): high doses (>100 mg per day of chlorpromazine) increase blood glucose levels (reduced insulin release).

Warn the patient and emphasise the importance of blood glucose monitoring. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with the neuroleptic agent.

- **Glucocorticoids** (systemic and local route: intra-articular, cutaneous and rectal preparations) and tetracosactrin: increase in blood glucose levels with possible ketosis (reduced tolerance to carbohydrates due to glucocorticoids).

Warn the patient and emphasise the importance of blood glucose monitoring, particularly at the start of treatment. It may be necessary to adjust the dose of the antidiabetic active substance during and after treatment with glucocorticoids.

- **Ritodrine, salbutamol, terbutaline: (I.V.)**

Increased blood glucose levels due to beta-2 agonist effects.

Emphasise the importance of monitoring blood glucose levels. If necessary, switch to insulin.

3) Combination which must be taken into account

- **Anticoagulant therapy** (Warfarin ...): Sulphonylureas may lead to potentiation of anticoagulation during concurrent treatment.

Adjustment of the anticoagulant may be necessary.

PREGNANCY AND LACTATION:

Pregnancy:

Uncontrolled diabetes (gestational or not) is associated with a higher incidence of congenital abnormalities and perinatal mortality. Blood glucose control should be optimal around the time of conception to reduce the risk of congenital malformations. It is recommended that insulin be used during pregnancy in diabetic women. Although no teratogenic effect has been reported in animal studies, gliclazide should not be administered during pregnancy.

Lactation:

Some sulphonylurea drugs are excreted in human milk although it is not known whether gliclazide is one of them. Because the potential for hypoglycemia in nursing infants may exist, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

EFFECTS ON ABILITY TO DRIVE & USE MACHINE:

This medicine has no known influence on the ability to drive and use machines. However, patients should be made aware of the symptoms of hypoglycaemia and should be careful if driving or operating machinery, especially at the beginning of treatment.

ADVERSE REACTIONS:

Based on the experience with gliclazide and with other sulphonylureas, the following undesirable effects have to be mentioned.

Hypoglycaemia

As for other sulphonylureas, treatment with Sun-Glizide 30MR Tablet can cause hypoglycaemia, if mealtimes are irregular and, in particular, if meals are skipped. Possible symptoms of hypoglycaemia are: headache, intense hunger, nausea, vomiting, lassitude, sleep disorders, agitation, aggression, poor concentration, reduced awareness and slowed reactions, depression, confusion, visual and speech disorders, aphasia, tremor, paresis, sensory disorders, dizziness, feeling of powerlessness, loss of self-control, delirium, convulsions, shallow respiration, bradycardia, drowsiness and loss of consciousness, possibly resulting in coma and lethal outcome.

In addition, signs of adrenergic counter-regulation may be observed: sweating, clammy skin, anxiety, tachycardia, hypertension, palpitations, angina pectoris and cardiac arrhythmia.

Usually, symptoms disappear after intake of carbohydrates (sugar). However, artificial sweeteners have no effect. Experience with other sulphonylureas shows that hypoglycaemia can recur even when measures prove effective initially.

If a hypoglycaemic episode is severe or prolonged, and even if it is temporarily controlled by intake of sugar, immediate medical treatment or even hospitalisation are required.

Gastrointestinal disturbances, including abdominal pain, nausea, vomiting dyspepsia, diarrhoea, and constipation have been reported: if these should occur they can be avoided or minimised if gliclazide is taken with breakfast.

The following undesirable effects have been more rarely reported:

- Skin and subcutaneous tissue disorders: rash, pruritus, urticaria, erythema, maculopapular rashes, bullous reactions.
- Blood and lymphatic system disorders: Changes in haematology are rare. They may include anaemia, leucopenia, thrombocytopenia, granulocytopenia. These are in general reversible upon discontinuation of medication.
- Hepato-biliary disorders: raised hepatic enzyme levels (AST, ALT, alkaline phosphatase), hepatitis (isolated reports). Discontinue treatment if cholestatic jaundice appears.

These symptoms usually disappear after discontinuation of treatment.

- Eye disorders
Transient visual disturbances may occur especially on initiation of treatment, due to changes in blood glucose levels.
- Class attribution effects:
Cases of erythrocytopenia, agranulocytosis, haemolytic anaemia, pancytopenia and allergic vasculitis, have been described for other sulphonylureas.

OVERDOSAGE:

Symptoms: Overdosage with sulfonylureas may result in hypoglycemia but it should be noted that the dosage which causes such hypoglycemia varies widely and may be within the accepted therapeutic range in sensitive individuals.

The manifestations of hypoglycemia include sweating, flushing or pallor, numbness, chilliness, hunger, trembling, headache, dizziness, increased pulse rate, palpitations, increased blood pressure and apprehensiveness in mild cases. In more severe cases, coma appears.

However, symptoms of hypoglycemia are not necessarily as typical as those described above and sulfonylureas may cause insidious development of symptoms mimicking cerebrovascular insufficiency.

Treatment: Discontinue medication and treat hypoglycemia by giving dextrose promptly and in sufficient quantity.

Some sulfonylurea-induced hypoglycemias may be refractory to treatment and susceptible to relapse especially in elderly or malnourished patients. Continuous dextrose infusions for hours or days have been necessary.

DOSAGE AND ADMINISTRATION:

Route of administration: Oral.

The usual daily dosage is 1 to 4 tablets per day as a single dose. The tablet must be taken whole with half a glass of water preferably just before breakfast. Tablet administration must always be followed by a meal in order to avoid or to minimise the possible onset of digestive disorders.

In all cases, strictly comply with doctor's prescription.

Do not take a double dose to compensate for a forgotten dose.

SHELF LIFE:

2 Years

STORAGE CONDITION:

Store below 25°C and away from excess heat and moisture.

PACKAGE:

30 Tablets packed in a sheet of PTP package. 2 sheets of the PTP package are packaged in a paper box.

30 Tablets packed in a sheet of PTP package. 3 sheets of the PTP package are packaged in a paper box.

30 Tablets packed in a sheet of PTP package. 4 sheets of the PTP package are packaged in a paper box.

PRODUCT DESCRIPTION:

It is white to off-white, oblong tablet, with scored 427 on one side and ST on the other side.

REGISTRATION NO.: MAL11030077AZ

Manufacturer:

TAIWAN BIOTECH CO., LTD.

22, Chieh Shou Road, Taoyuan, Taiwan, R.O.C.

Product Registration Holder:

SUNWARD PHARMACEUTICAL SDN BHD

No. 9,11,17, Jalan Kempas 4, Taman Perindustrian Tampoi Indah, 81200 Johor Bahru, Johor, Malaysia.

Revised Date: 17 Apr 2025